LRS31 Asset Pricing and Portfolio Theory

Home Assignment 3

Use RStudio to complete all numerical calculations, data processing, statistical tests, portfolio optimisation, and plots in R. Show analytical derivations where requested. Submit your R script (`.R`) or R Markdown/Quarto source together with your written answers and relevant R output. Your code must reproduce your reported results.

Use the Monthly worksheet from HA2. Calculate simple monthly returns from the five stocks' total-return indices, using decimals in R and reporting returns and volatilities in percent. Use sample variances and covariances with denominator $T-1$, and annualise mean monthly returns as $(1+\bar{r})^{12}-1$.

Use January 2011–December 2021 returns for Q1–Q3, a fixed 132-month rolling estimation window for Q4, and January–September 2022 for Q5–Q6. Retain December 2010 observations where needed. Use preceding month-end market capitalisations for value weights. For Q3–Q6, use only information available before each investment month when estimating portfolio weights. Short selling and risk-free borrowing are permitted unless otherwise stated.

1. Calculate the value-weighted portfolio return as in HA2 Q6, now subject to a maximum weight of 35% in any single asset. Briefly explain how you enforce the weight limit. What is the mean realised return per annum?

a. Less than 24%
b. 24–27%
c. 27–30%
d. More than 30%

2. HA2 Q10 continued. Calculate the global minimum-variance portfolio's weights, expected return, and volatility. **Answer:** What is its expected return per annum?

a. Less than 18%
b. 18–19%
c. 19–20%
d. More than 20%

3. Assume that the risk-free rate is given in column AQ. Divide by 100 to convert the annual percentage rate to a decimal, and then divide by 12 to approximate the monthly rate. Use the rate observed at the end of the preceding month: for January 2022, use the value from 31 December 2021; for February, use the value from the end of January; and so on. This ensures that the risk-free return is known before the investment period.

Re-estimate the return and volatility of the portfolio on the capital market line with a target return of 1% per month. What is the expected volatility for January 2022?

a. Less than 1.8%
b. 1.8–2.0%
c. 2.0–2.2%
d. More than 2.2%

4. **(2 points) Q3 continued.** Continue in a rolling fashion for February–October 2022. Each month, add the newly available observation and drop the oldest observation, keeping the estimation-window length fixed. Estimate the expected returns and variance–covariance matrix, then calculate the expected portfolio volatility for each month. **Answer:** What is the average expected monthly volatility over January–October 2022, including the January estimate from Q3?

   R hint: use `colMeans()` and `cov()` within a rolling-window loop. If calculating the sample covariance matrix directly, let $\mathbf{R}$ be the $T \times N$ return matrix and $\bar{\mathbf{r}}^\top$ the $1 \times N$ row vector of column means. Then

$$\mathbf{R}_e = \mathbf{R} - \mathbf{1}_T \bar{\mathbf{r}}^\top, \qquad \widehat{\mathbf{\Sigma}} = \frac{\mathbf{R}_e^\top \mathbf{R}_e}{T-1}.$$

   a. Less than 1.8%
   b. 1.8–2.0%
   c. 2.0–2.2%
   d. More than 2.2%

5. Use the rolling portfolio weights from Q4, including the January weights from Q3, to calculate the realised portfolio return for January–September 2022. Report the cumulative realised return over that period.

   a. Less than 15%
   b. 15–20%
   c. 20–25%
   d. More than 25%

6. **(2 points)** Calculate the Sharpe ratio for the modern portfolio theory (MPT) portfolio in Q5 using monthly data for January–September 2022. Also calculate mean excess returns and volatilities for the rolling minimum-variance portfolio (estimated as in Q4), the equally weighted portfolio, and the value-weighted portfolio. These three comparison portfolios do not include the risk-free asset. Rebalance the equally weighted portfolio monthly and use uncapped value weights for the value-weighted portfolio.

   Calculate their Sharpe ratios for January–September 2022. Which of the four strategies has the highest Sharpe ratio?

   a. MPT portfolio
   b. Equally weighted portfolio
   c. Value-weighted portfolio
   d. Minimum-variance portfolio